A CHARACTERISATION OF ± 1-FRAMED UNKNOTS

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Abstract. We give a new, diagrammatic, proof, involving virtual knots, that if the quotient of the fundamental group of a blackboard-framed, oriented knot by the normal closure of its longitude is trivial, then it is a ± 1-framed unknot. The arguments make use only of Reidemeister moves (no Kirby moves), providing new proofs of several important theorems concerning the 3-sphere.

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1. Introduction

Questions about 3-manifolds can sometimes be reformulated entirely in terms of diagrammatic knot theory, by means of the bridge between the two fields provided by Dehn surgery and Kirby's theorem. In this article, we shall give a new proof of the following (the definition of longitude with which we shall work will be recalled at the beginning of §2; technically it is defined with respect to a chosen point of K). We shall always speak of the fundamental group or rack of an oriented (classical or virtual) knot diagram, as opposed to of a knot itself, thinking of this group or rack as defined directly from the diagram using Wirtinger relations.

Theorem 1. *Let K be an oriented (classical) knot diagram, corresponding to a knot equipped with its blackboard framing. If $\pi_1(K)/\langle l \rangle$ is trivial, where $\langle l \rangle$ is the normal closure in $\pi_1(K)$ of the longitude l of K, then K is equivalent under R2 and R3 moves to a ± 1-framed unknot (see Figure 1).*

Though we will not go into the details of this, Theorem 1 immediately implies, via the afore-mentioned bridge, a version of the Property P conjecture: if a 3-manifold obtained by non-trivial integral Dehn surgery on a knot has trivial fundamental group, then this knot is the unknot. This can also be formulated as follows, sometimes referred to as the Poincaré conjecture for knots: if a 3-manifold obtained from a knot by integral Dehn surgery has trivial fundamental group, then it is the 3-sphere (it is only the ± 1-framing of the unknot that corresponds to the 3-sphere under Dehn surgery). In addition, Theorem 1 implies the Gordon–Luecke

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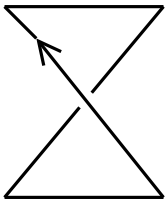


FIGURE 1. A -1-framed unknot. In a $+1$-framed unknot, the over-arc and under-arcs of the crossing are switched.

theorem that if an integer Dehn surgery on a knot yields the 3-sphere, then the knot is equivalent to the unknot. All of this was already known ([7, 8, 10, 9, 4]), but the proofs are very different to the one we shall give of Theorem 1.

Let w be any word belonging to the free group on the arcs of K whose equivalence class in $\pi_1(K)/\langle l \rangle$ is the empty word. Our approach to Theorem 1 involves showing that if w is not already trivial in $\pi_1(K)$, then one can construct an oriented virtual knot diagram V, into which the over-arcs of K inject, whose longitude (up to this injection) is exactly w. This is Lemma 4.

Diagrammatic approaches to Theorem 1 have been attempted before, but are notoriously thorny. There is no simple way to achieve a construction as just described using classical knot diagrams, because it is not necessarily local; a sequence of Reidemeister moves to adjust part of the longitude might have consequences for other parts of it. Passing to virtual knots allows us to entirely avoid intricacies of this kind.

A specific ingredient in our proof of Lemma 4 is what appears to be a new notion of n-parallelisation to a virtual knot of a classical knot. In the paradigmatic case, it has the property that its longitude is the product of the longitude of the classical knot with itself n times.

We observe that if an oriented virtual knot diagram has a longitude which cancels to a single arc or its formal inverse in the free group on the arcs of this diagram, then its fundamental rack is trivial. This is Lemma 6. To conclude the proof of Theorem 1, we combine Lemma 4 and Lemma 6 in a straightforward way, relying on the fact that the fundamental rack is a complete invariant of classical knots.

In §3, we indicate directions in which the techniques we introduce in this article might be further explored.

2. PROOF

Let K be as in Theorem 1. Fix a point p on an arc of K. Recall that when working with its blackboard framing, the longitude l of K starting at p can be obtained by beginning at p with the empty word, then walking around K in accordance with its orientation, appending a (resp a^{-1}) whenever one passes under a crossing whose over-arc is a, and where the orientation of the arcs is as on the left (resp right) in Figure 2. The opposite sign convention is sometimes used. The choice of p makes no difference to $\pi_1(K)/\langle l \rangle$ up to isomorphism.

The longitude of an oriented virtual knot diagram is defined [6] similarly: one simply walks through any virtual crossings one encounters without appending anything, ie only classical crossings contribute.

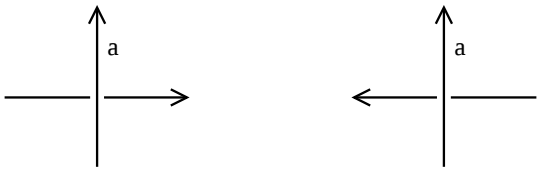


FIGURE 2. Longitude orientations

We shall work with labellings of the arcs of oriented virtual link diagrams. Whenever we apply a Reidemeister move to a certain diagram V, we intend that the labelling of its arcs be adjusted in the standard way. In the case, for instance, of a classical R2 move of a certain arc labelled a under another arc, the arc labelled a is split into three, the first and last of which (when we travel in accordance with the orientation of V) retain the label a, whilst the second is given a new label that was not previously used.

As ever, the importance of this is that the labelling of the post-move diagram defines a presentation of its fundamental group and fundamental rack. The generators of this presentation are the set of labels of the post-move diagram, whilst the relations are the Wirtinger relations coming from each crossing of the post-move diagram.

We shall make use of the following construction. The author is not aware that it has previously been considered; a notion of parallelisation somewhat akin to it was introduced in a recent article [5], but it is not exactly the same. Figure 3 gives an example of the construction, for a trefoil; in this case, no orientations have been reversed in step 3, and no moves applied in step 4.

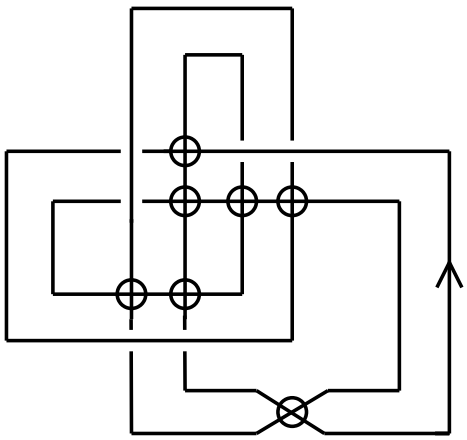


FIGURE 3. Doubling/2-parallelisation of a trefoil

Definition. Let K be an oriented classical knot diagram, and p a point on an arc of it. An *n-parallelisation* of K at p, where $n \geq 1$, is an oriented virtual knot diagram obtained as follows.

1. Starting at p, walk around K in accordance with its orientation, adding a parallel path as one goes, slightly offset from the original path. Wherever the original path crosses classically under an arc of K, ensure that the parallel path does the same. Wherever other crossings are necessary, make these virtual, including where the original path crosses classically over another arc. See Figure 4 for the various possibilities.

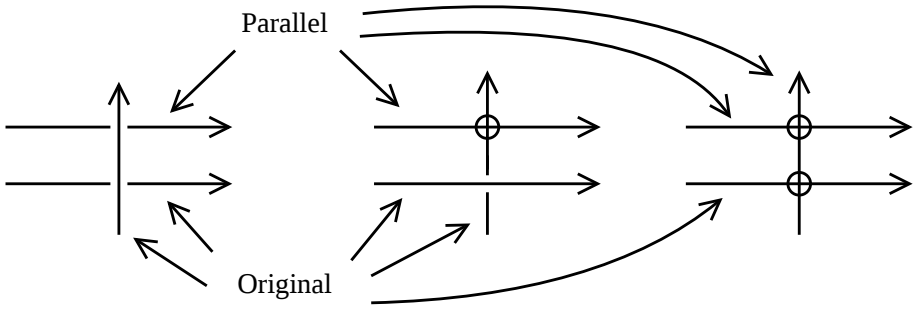


FIGURE 4. Crossings involving a parallel track

2. Repeat this procedure until one has n parallel paths in total (including the original one).
3. Reverse the orientations of any number of the n parallel paths.
4. Apply a finite sequence of moves, each of which is either a classical R2 or R3 move or a virtual R1, R2, or R3 move (as defined in [6], in all three cases), to the oriented virtual link diagram obtained after step 3. For every component of the latter, we require that a small neighbourhood of the copy of p belonging to it be fixed by all of these moves.
5. From each component of the oriented virtual link diagram obtained after step 4, remove a small segment of the arc containing the copy of p. Each component can then be thought of as having a start and an end (which is which being determined by the orientation of the component).

   For $1 \leq i < n$ (resp $i = n$), join, without introducing any classical crossings, the end of the i-th component to the start of the $(i+1)$-th (resp first) component, where i refers to the order in which the components were added in steps 1 and 2. If crossings are necessary, make these virtual. We refer to the arc constructed in this way as the *i-th joining arc* of the resulting oriented knot diagram.

   Some examples are given in Figure 5. To avoid clutter, these are simplified somewhat: there may be other arcs intersecting the empty region in the middle of the left-hand sides of these examples, and virtual crossings of these might then be necessary on their right-hand sides.

The following notion will play a central role in this article.

Definition. Let K be an oriented classical knot diagram, and p a point on an arc of K. We shall say that a word $w = a_1^{\pm 1} a_2^{\pm 1} \cdots a_n^{\pm 1}$ in the free group on the arcs of K is *completely realisable from p* if there is an m-parallelisation V of K at p, for

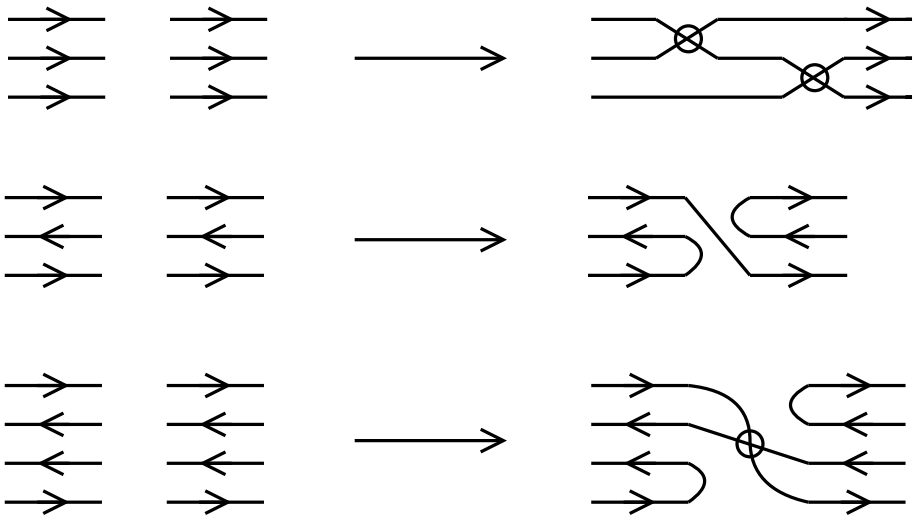


FIGURE 5. Joining parallel tracks

some $m \geq 1$, such that the longitude of V starting at any point on the m-th joining arc of V is equal (on the nose) to w.

Here $a_1, \ldots, a_n$, as well as being labels of arcs of K, are to be taken to be labels of the corresponding arcs in the copy of K with which step 1 of the construction of V begins. These labels are to be followed through the other steps in the construction of V, it being required that they survive step 4 of the construction, ie, for each $1 \leq i \leq n$, there is at least one arc of the oriented virtual link diagram L which one obtains after step 4 which has label a_i. If the arc to which p belongs on the first component of L (ie the copy of K with which we began in step 1) has label a_i for some $1 \leq i \leq n$, both the first joining arc of V and the m-th joining arc of V are to be labelled a_i. All arcs of V which do not acquire a label in one of these ways may be labelled freely.

Let C be a classical crossing of an oriented virtual link diagram V. We shall throughout let w_C denote the word $c^{-1}b^{-1}ab$ in the free group on the arcs of V if C has the orientation on the left in Figure 6, but let w_C denote the word $c^{-1}bab^{-1}$ if C has the orientation on the right in Figure 6.

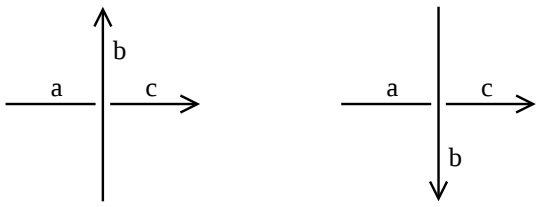


FIGURE 6. Crossing orientations

Lemma 2. *Let K be an oriented classical knot diagram, and p a point on an arc of K. Let w_1 and w_2 be words (either or both of which may be empty) in the free group on the arcs of K, such that $w_1 w_2$ is completely realisable from p. For any crossing C of K, and any word g in the free group on the arcs of K, the word*

$$w_1 \left(g^{-1} w_C^{\pm 1} g\right) w_2$$

in the free group on the arcs of K is completely realisable from p.

Proof. Let $a_1, \ldots, a_n$ be arcs of K such that $g = a_1^{\pm 1} \cdots a_n^{\pm 1}$. Let V be an m-parallelisation of K at p, for some $m \geq 1$, whose longitude starting at any point p' on the m-th joining arc of V is $w_1 w_2$, identifying labels of arcs of K with labels of arcs of V in the manner explained in the definition of complete realisability. Let L be the oriented virtual link diagram obtained after step 4 of the construction of V. Again, identify labels of arcs of K with labels of arcs of L as in the definition of complete realisability.

Suppose first that C is oriented as on the left in Figure 6, and that the exponent of w_C (in the statement of the lemma) and of the a_1 term in g is $+1$ in both cases. Using (only) virtual R2 moves, along with a single virtual R1 move if necessary, drag a small segment of the arc of L labelled a_1 so that it lies as in Figure 7 in relation to C.

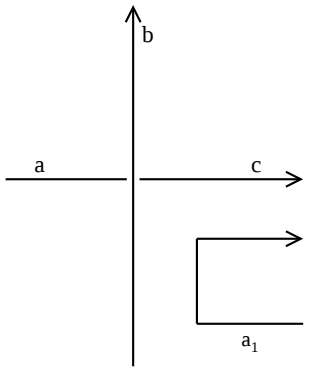


FIGURE 7. Dragging a segment of a_1.

If w_1 is not the empty word, let C' be the crossing of L corresponding to the final crossing of V which contributes a term to w_1 when constructing the longitude of V starting at p'. Take a small segment of the arc of L upon which one emerges having passed under C', ensuring, if that arc contains p', that this segment lies strictly between C' and p'. If w_1 is instead the empty word, take rather a small segment of the arc of the first component of L (in the order in which the components of L were added during its construction) to which p' belongs, ensuring that this segment lies strictly between p' and the first crossing of that component of L.

Using (only) virtual R2 moves, along with a single virtual R1 move if necessary, drag this segment so that it is positioned as in Figure 8 in relation to C. Using three classical R2 moves followed by a classical R3 move, drag this segment further, under C, to obtain Figure 9.

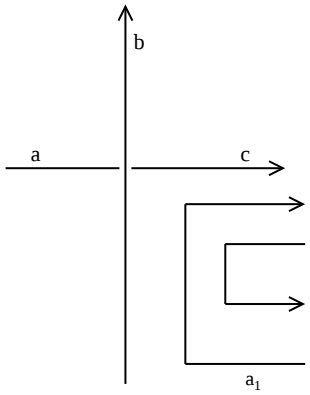


FIGURE 8. Dragging a segment of the arc upon which emerges having crossed under C'

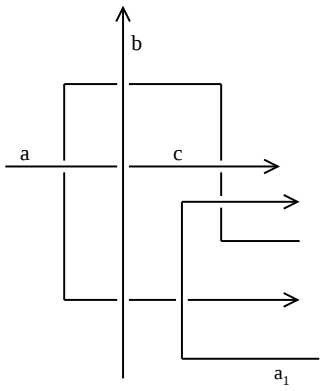


FIGURE 9. Dragging under C

Replacing L by the oriented virtual link diagram we have constructed to this point, we obtain an m-parallelisation of K at p whose longitude starting at p' is

$$w_1 \left(a_1^{-1} c^{-1} b^{-1} a b a_1\right) w_2.$$

If instead the exponent of the a_1 term in g is -1, proceed in exactly the same way, except that the dragged segment of a_1 should (using a virtual R1 move if necessary) have the opposite orientation, as in Figure 10.

Suppose now that the exponent of the a_2 term in g is $+1$. To obtain an m-parallelisation of K at p whose longitude starting at p' is

$$w_1 \left(a_2^{-1} a_1^{-1} c^{-1} b^{-1} a b a_1 a_2\right) w_2,$$

proceed exactly as above, but in addition drag a segment of the arc a_2 so that it nestles inside the dragged segment of a_1, as in Figure 11 when the exponent of the a_1 term in g is also $+1$. When the exponent of the a_1 (resp a_2) term in g is -1, the arc a_1 (resp a_2) should (using a virtual R1 move if necessary) have the opposite orientation to the one it has in Figure 11.

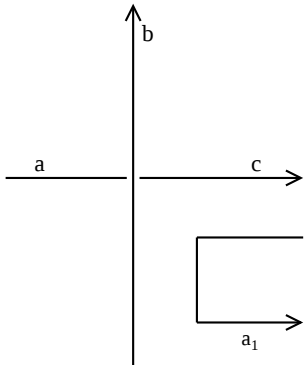


FIGURE 10. Dragging a_1 so that it has the opposite orientation

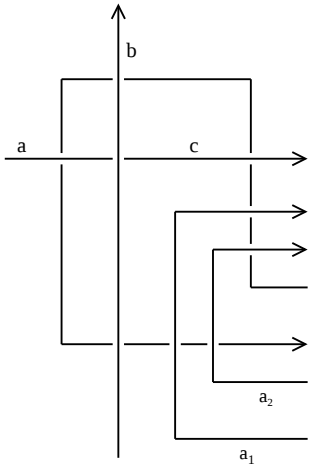


FIGURE 11. With both a_1 and a_2

Carrying out the corresponding sequences of moves for $a_3, \dots, a_n$, we obtain an m-parallelisation of K at p whose longitude starting at p' is

$$w_1 \left(g^{-1} w_C^{\pm 1} g\right) w_2,$$

as required.

When the exponent of w_C in the statement of the lemma is still $+1$, but C is instead as on the right in Figure 6, an entirely analogous argument can be given. For example, Figure 12 is the required replacement for Figure 9 when exhibiting that there is an m-parallelisation of K at p whose longitude starting at p' is

$$w_1 \left(a_1^{-1} c^{-1} b a b^{-1} a_1\right) w_2.$$

The same is true when the exponent of w_C in the statement of the lemma is -1, and C is of either of the two kinds in Figure 6. In this case, the segment of arc

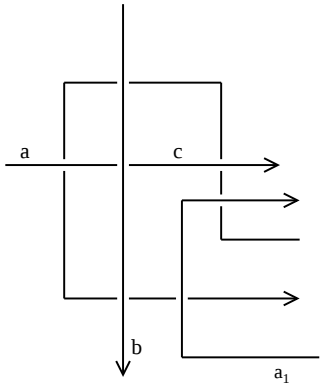


FIGURE 12. Dragging under C when b has the opposite orientation

which is dragged to obtain Figure 8 should have the opposite orientation, using a virtual R1 move if necessary. Figure 13 is, for instance, the required replacement for Figure 9 when exhibiting that there is an m-parallelisation of K at p whose longitude starting at p' is

$$w_1 \left(a_1^{-1} b^{-1} a^{-1} b c a_1\right) w_2.$$

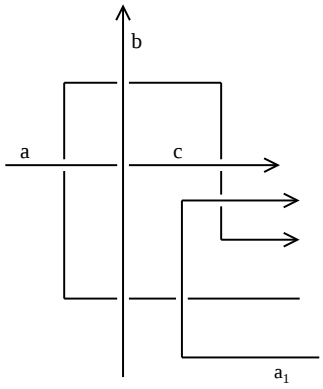


FIGURE 13. Dragging under C when w_C has exponent -1

□

We shall also require the following.

Lemma 3. *Let K be an oriented classical knot diagram, and p a point on an arc of K. Let l be the longitude of K starting at p. For all $m \geq 1$, every word*

$$g_1^{-1} l^{\epsilon_1} g_1 \cdots g_m^{-1} l^{\epsilon_m} g_m$$

in the free group on the arcs of K is completely realisable from p, where, for all $1 \leq i \leq m$, we have that $\epsilon_i = \pm 1$ and that $g_i = a_{i,1}^{\pm 1} \cdots a_{i,n_i}^{\pm 1}$ for certain arcs $a_{i,1}, \ldots, a_{i,n_i}$ of K.

Proof. For the integer m in the statement of the lemma, carry out steps 1–3 of the construction of an m-parallelisation of K. Let L denote the resulting oriented virtual link diagram. In step 3, reverse the orientation of the i-th component of L if and only if $\epsilon_i = -1$.

For a given i, suppose that the exponent of the $a_{i,1}$ term of g_i is $+1$. Using (only) virtual R2 moves, along with a virtual R1 move if necessary, drag a segment of the arc labelled $a_{i,1}$ of the first component of L so that it is near the copy p' of p which belongs to the i-th component of L (where the indexing of the components of L refers to the order in which they were added in steps 1 and 2). See Figure 14.

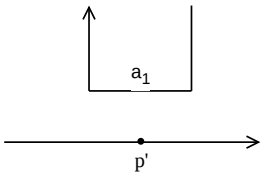


FIGURE 14. Dragging a_1

Next, apply a classical R2 move so as to obtain Figure 15.

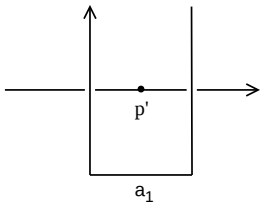


FIGURE 15. After an R2 move

If instead the exponent of the $a_{i,1}$ term of g_i is -1, proceed entirely analogously, except that the segment which is dragged should, using a virtual R1 move if necessary, have the opposite orientation near p'. See Figure 16.

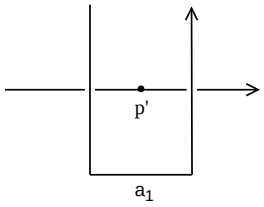


FIGURE 16. With the opposite orientation of a_1

Next, drag a segment of the arc labelled $a_{i,2}$ of the first component of L near to p' using only virtual R2 moves, and if necessary a virtual R1 move, so that it nestles inside the dragged segment of the arc labelled $a_{i,1}$, and then apply a classical R2 move so as to obtain either Figure 17 in the case that the exponent of the $a_{i,2}$

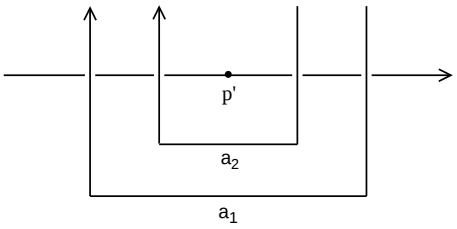


FIGURE 17. With a_2 as well

term of g_i is $+1$, or the same but with $a_{i,2}$ having the opposite orientation if the exponent of the $a_{i,2}$ term of g_i is -1.

Proceed entirely analogously for $a_{i,3}, \dots, a_{i,n_i}$. Let V be the m-parallelisation of K at p for which step 4 consists exactly of these sequences of moves for all $1 \leq i \leq m$. The longitude of V starting at any point on the m-th joining arc of V is exactly

$$g_1^{-1} l^{\epsilon_1} g_1 \cdots g_m^{-1} l^{\epsilon_m} g_m,$$

as required. □

We deduce the following.

Lemma 4. *Let K be an oriented classical knot diagram, and p a point on an arc of K. Let l be the longitude of K starting at p. Let w be a word*

$$g_1^{-1} w_1 g_1 \cdots g_m^{-1} w_m g_m$$

in the free group on the arcs of K, where $m \geq 1$. For all $1 \leq i \leq m$, suppose that:

i) the word g_i is as in the statement of Lemma 3;
ii) the word w_i is equal either to $l^{\pm 1}$ or to $w_C^{\pm 1}$, for a classical crossing C of K.

Suppose moreover that w_i is equal to $l^{\pm 1}$ for at least one i. Then w is completely realisable from p.

Proof. Let $1 \leq i_1, \dots, i_r \leq m$ be those indices for which w_{i_j} is equal to $l^{\pm 1}$. By Lemma 3, every word

$$g_{i_1}^{-1} l^{\epsilon_{i_1}} g_{i_1} \cdots g_{i_r}^{-1} l^{\epsilon_{i_r}} g_{i_r}$$

in the free group on the arcs of K is completely realisable from p. The lemma follows immediately from this by repeated application of Lemma 2. □

Given a rack R, recall that there are maps $\triangleleft\colon R \times R \to R$ and $\triangleright\colon R \times R \to R$ such that

$$(a_1 \triangleleft a_2) \triangleright a_2 = (a_1 \triangleright a_2) \triangleleft a_2 = a_1$$

for all $a_1, a_2 \in R$. Sometimes both $\triangleleft$ and $\triangleright$ are part of the definition of R, whilst sometimes only one of them is, in which case the other is derived from it. We shall sometimes write $\triangleleft^\epsilon$, where $\epsilon = \pm 1$, in which case $\triangleleft^1$ should be taken to be $\triangleleft$, and $\triangleleft^{-1}$ to be $\triangleright$.

Lemma 5. *Let $\left(R, \triangleleft^{\pm 1}\right)$ be a rack with a presentation consisting of a set G of generators and, for some $n \geq 1$, exactly n relations of the following form:*

i) $x_i \triangleleft^{\epsilon_i} a_i = x_{i+1}$ for $1 \leq i \leq n-1$;

ii) $x_n \triangleleft^{\epsilon_n} a_n = x_1$.

Here $x_i \in G$, $a_i \in G$, *and* $\epsilon_i = \pm 1$.

Suppose that every element of G *appears as* x_i *or* a_i *in at least one of these relations. Suppose moreover that there is an odd integer* m, *with* $1 \leq m \leq n$, *such that, for all* $1 \leq i \leq n$ *with* $i \neq m$, *there is an integer* $j_i \geq 1$ *for which either:*

a) $a_{i+k} = a_{i+2j_i-1-k}$ *and* $\epsilon_{i+k} = -\epsilon_{i+2j_i-1-k}$ *for all* $0 \leq k \leq j_i - 1$*; or*
b) $a_{i-k} = a_{i-2j_i+1+k}$ *and* $\epsilon_{i-k} = -\epsilon_{i-2j_i+1+k}$ *for all* $0 \leq k \leq j_i - 1$.

Then R *is the unique rack with a single element.*

Proof. If R has a single generator, the only relations satisfying the hypotheses of the lemma are $x \triangleleft x = x$ and $x \triangleright x = x$. Either of these relations immediately implies that x is the only element of R, as required.

Otherwise, let a be any $a_i \in G$ for which $i \neq m$. It follows immediately from a) and b) that either $a_{i+j_i-1} = a_{i+j_i}$ and $\epsilon_{i+j_i-1} = -\epsilon_{i+j_i}$, or else $a_{i-j_i} = a_{i-j_i+1}$ and $\epsilon_{i-j_i} = -\epsilon_{i-j_i+1}$.

Suppose that the first of these two possibilities holds. By i) and ii) in the hypotheses of the lemma, we then have that $x_r \triangleleft^{\epsilon_r} a_r = x_{r+1}$ and that $x_{r+1} \triangleleft^{-\epsilon_r} a_r = x_{r+2}$, where $r = i + j_i - 1$.

These two relations imply that $x_r = x_{r+2}$. For if $\epsilon_r = 1$, we have that

$$x_r = (x_r \triangleleft a_r) \triangleright a_r = x_{r+1} \triangleright a_r = x_{r+2},$$

and similarly if $\epsilon_r = -1$ (interchange $\triangleleft$ and $\triangleright$).

Thus we can replace our original presentation of R by one in which, firstly, we remove x_{r+2} as a generator of the presentation of R; secondly, we remove the aforementioned two relations involving x_r; and, thirdly, we replace x_{r+2} by x_r in all of the other relations in i) and ii) which involve x_{r+2}.

After re-indexing, the hypotheses i), ii), a), and b) are all still satisfied by our new presentation of R. Since this presentation has strictly fewer generators than the original one, the lemma holds for it by induction.

An entirely analogous argument can be given if the second possibility described in the second paragraph of the proof holds.

□

We deduce the following. Recall that the fundamental rack of an oriented virtual link diagram is defined as for oriented classical link diagrams, simply ignoring any virtual crossings.

Lemma 6. *Let* V *be an oriented virtual knot diagram. Suppose that there is an arc* b *of* V *for which the longitude* l *of* V*, starting at a certain point, is equal in the free group on the arcs of* V *to* $b^{\pm 1}$. *Then the fundamental rack of* V *is the unique rack with a single element.*

Proof. The hypotheses a) and b) of Lemma 5 hold for the fundamental rack of any oriented virtual link diagram. Since, by the hypotheses of the lemma, it is possible to obtain $b^{\pm 1}$ by cancelling sub-words of l of the form $a^{\pm 1}a^{\mp 1}$, where a is an arc of V, the hypotheses i) and ii) of Lemma 5 also hold for the fundamental rack of V. That the lemma holds thus follows immediately from Lemma 5. □

We shall also require the following.

Lemma 7. *Let K be an oriented classical knot diagram, and p a point on an arc of K. For some $n \geq 1$, let V be an n-parallelisation of K at p. The fundamental rack of K is a quotient of the fundamental rack of V.*

Proof. When $n = 1$, that V is an n-parallelisation of K is the same as to say that there is a sequence of moves exhibiting V to be equivalent to K, each of which is either a classical R2 or R3 move or a virtual R1, R2, or R3 move. Thus the lemma in this case follows immediately from the fact that fundamental racks of oriented virtual link diagrams are invariant up to isomorphism under all of these moves [1].

Suppose that $n > 1$. Let L (resp L') be the oriented virtual link diagram obtained after step 3 (resp step 4) of the construction of V. For $1 \leq i \leq n$, let a_i and a_i' be the arcs of L and L' respectively which contain the copy of p belonging to their i-th components (where i, here and throughout the proof, refers to the order in which the components were added to L in steps 1 and 2 of the construction of V). Let a_i'' be the i-th joining arc of V.

Let Q'' (resp Q' and Q) be the quotient of the fundamental rack of V (resp L' and L) obtained by identifying a_i'' (resp a_i' and a_i) with a_1'' (resp a_1' and a_1) for all $1 \leq i \leq n$. By definition, V and L' are identical except that the arcs of L' which contain a copy of p are replaced by the joining arcs of V. Since the arcs of L' which contain a copy of p (resp the joining arcs of V) are all identified with one another in Q' (resp Q''), we deduce that Q'' is isomorphic to Q'.

Since the fundamental rack of any oriented virtual link diagram is invariant up to isomorphism under classical R2 and R3 moves and under virtual R1, R2, and R3 moves, we observe moreover that Q' is isomorphic to Q. Hence Q'' is isomorphic to Q.

Fix a given $1 \leq i \leq n$. Walk around the i-th component of L, starting at the copy of p which belongs to it. Proceed in accordance with the orientation of L if this is the same as the orientation of the first component of L, but in reverse if it is not.

Let b be the over-arc of the first classical crossing C of L which we pass under. Let x be the arc of the i-th component of L which we travel along having passed under C. The fundamental rack of L has $a_i \triangleright^{\epsilon} b = x$ as one of its relations, where ϵ is either $+1$ or -1 (depending upon the orientation of b). Since we identify a_i with a_1 in Q, it follows that $a_1 \triangleright^{\epsilon} b = x$ in Q.

However, let c be the arc belonging to the first component of L which we travel along having passed under b after walking along a_1 (in accordance with its orientation). We have that $a_1 \triangleright^{\epsilon} b = c$ in the fundamental rack of L, and hence in Q. We deduce that $x = c$ in Q.

Let C' be the next classical crossing of L which we pass under when walking around the i-th component of L. Let b' be the over-arc of C', and let x' be the arc of the i-th component of L which we travel along having passed under C'. We have that $x \triangleright^{\epsilon'} b' = x'$ in the fundamental rack of L, where $\epsilon' = \pm 1$. Hence $c \triangleright^{\epsilon'} b' = x'$ in Q. As above, letting c' be the arc of the first component of L which we travel along having passed under b' after walking along c, we have that $c \triangleright^{\epsilon'} b' = c'$ in the fundamental rack of L, and hence that $x' = c'$ in Q.

Proceeding in this way, walking around the entirety of the i-th component of L, we deduce that every arc belonging to the i-th component of L is equal in Q to an arc belonging to the first component of L. Since this holds for all $1 \leq i \leq n$, every relation of the Wirtinger presentation of the fundamental rack of L in which the

three involved arcs do not all belong to the first component of L can be discarded in Q.

Since the first component of L is simply a copy of K, we deduce that, up to a bijection between their generators, Q admits a presentation which is identical to the Wirtinger presentation of the fundamental rack of K. Thus Q is isomorphic to the fundamental rack of K, and hence so is Q''. The latter is a quotient of the fundamental rack of V, as required.

□

We now bring everything together.

Proof of Theorem 1. Since $\pi_1(K)/\langle l \rangle$ is trivial, every element of the free group on the arcs of K lies within the normal closure in this free group of the set of words

$$\{w_C \mid C \text{ is a crossing of } K\} \cup \{l\}.$$

There is at least one arc b of K which is not equal, in the free group on the arcs of K, to a product of conjugates of words of the form $w_C^{\pm 1}$ for C a classical crossing of K, since otherwise $\pi_1(K)$ would be trivial, which is impossible for a knot. For this arc b, we have that

$$b = g_1^{-1} w_1 g_1 \cdots g_n^{-1} w_n g_n,$$

in the free group on the arcs of K, where $g_1, \ldots, g_n$ and $w_1, \ldots, w_n$ satisfy the hypotheses of Lemma 4.

Let p be the point at which we start when constructing the longitude l of K. By Lemma 4, there is an m-parallelisation V of K at p, for some $m \geq 1$, whose longitude starting at any point on the m-th joining arc of V is equal (on the nose) to the right-hand side of the above equality. By Lemma 6, we deduce that the fundamental rack of V is the trivial rack (the unique rack with a single element). By Lemma 7, this implies that the fundamental rack of K is the trivial rack.

The fundamental rack of the (classical) ± 1-framed unknot is also the trivial rack. Since the fundamental rack is a complete invariant of framed classical knots [2], we deduce that K is equivalent under classical R2 and R3 moves to the ± 1-framed unknot, as required.

□

It is possible to give a proof of Theorem 1 in which one replaces the definition of a completely realisable word w by one involving virtual knot diagrams equivalent to a connected sum of copies of K with itself. One requires that the longitude of V starting at p is equal to w in the quotient of the free group on the arcs of this connected sum by the kernel of the fold morphism to the free group on the arcs of K.

The proof of Lemma 2 goes through essentially unchanged in this setting. In the proof of Lemma 3, one instead drags the arcs a_1, a_2, and so on over the entire copies of K involved in forming the connected sum, as in Figure 18.

In the statement and proof of Lemma 6, one replaces the fundamental rack of V with its quotient Q by the kernel of the fold morphism from the fundamental rack of V to the fundamental rack of K. One replaces Lemma 7 by the more or less immediate observation that Q is isomorphic to the fundamental rack of K.

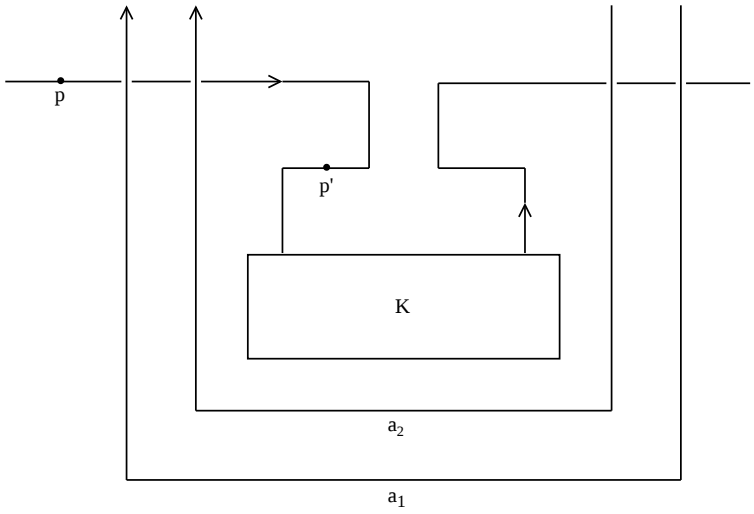


FIGURE 18. Dragging a_1 and a_2 over an entire copy of K

3. Further directions

The analogue of Theorem 1 for links should be provable along the same lines, but certain Kirby moves will be necessary, and aspects of the arguments we have given must be adjusted.

One might also try to replace the trivial group by other groups in the statement of Theorem 1. For the integers, one would aim to prove that K must be equivalent to the 0-framed unknot. This is the Property R conjecture, proven in a very different way in [3, 4]. For finite cyclic groups, the picture is more complicated, but in certain cases one would aim to prove that K is equivalent to a $\pm n$-framed unknot for some $n > 1$.

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